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Assignment 1

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Subject: Programming Fundamentals
Section: AB

Q1

1) #include <iostream>
using namespace std;
int main(){
int bon1, bon2, basicSal, finalSal, tax, years, rating;
cout << "Please enter your salary and experience";
cin >> salary;
cin >> years;
if (years < 2){
bon1 = basicSal * (5/100);
} else if (years >= 2 && years <= 5){
bon1 = basicSal * (10/100);
} else {
bon1 = basicSal * (15/100);
}
cout << "Please Enter your rating";
cin >> rating;
if (rating >= 9){
bon2 = basicSal * (5/100);
} else {
int totBon = bon2 = 0;
}
int totBon = bon1 + bon2;
basicSal = basicSal + totBon;

```
if ( totS basicSal > 1000000) {
        tax = basicSal * (10/100);
} else {
    tax = 0;
}
finalSal = basicSal - tax;
cout << "tax:" << tax << endl;
cout << "bonus:" << totBon << endl;
cout << "final salary:" << finalSal << endl;
if (rating >= 9) {
        cout << "Excellent performance";
} else if (rating >= 4 && rating <= 8) {
        cout << "Good performance";
} else {
    cout << "Needs Improvement";
}
return 0;
}
```

2)

```
#include <iostream>
using namespace std;
int main (){
        int units, basBill, finBill, surCharge, totBill;
        char userType;
        cout << "How much units did you consume?";
        cin >> units;
        if ( units <= 100){
                basBill = units * 10
        }
```

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else if (units>100 && units<=300) {
    basBill = units * 15;
  } else {
    basBill = units * 20;
  }
if (basBill>5000) {
  surCharge = basBill * (10/100);
  }
  else {
  surCharge = 0;
  }
  finBill = basBill + surCharge;
  cout << "Please enter d for domestic user and
          c for commercial user";
  cin >> userType;
  cout << "Units: " << units << endl;
  cout << "basic amount: " << basBill << endl;
  cout << "Surcharge: " << surCharge << endl;
  if (userType == 'D' || userType == 'd') {
      cout << "Thank you for conserving energy";
  } else {
      cout << "Commercial user";
  }
return 0;
}
```

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3)
```
#include <iostream>
using namespace std;
int main(){
    int mark1, mark2, mark3;
    double average;
    char grade;
    cout << "Please enter your marks";
    cin >> mark1;
    cin >> mark2;
    cin >> mark3;
    average = (mark1+mark2+mark3)/3;
    cout << average;
    if (average >= 85){
        grade = 'A';
        cout << grade << ", Excellent result";
    }else if (average >= 70 && average <= 85){
        grade = 'B' << " Very good";
        cout << grade << ", Very good";
    } else if (average >= 50 && average < 70){
        grade = 'C';
        cout << grade << ", Needs more effort";
    } else (average < 50 && average >= 40){
        grade = 'F';
        cout << grade << ", Work harder next time";
    } else{
    if(mark1 < 40 || mark2 < 40 || mark3 < 40){
        cout << "Supplementary Exam required";
    }
    return 0;
}
```

```
int main(){
    int arr[10];
```

4)

```
#include<iostream>
using namespace std;
int main(){
    int purPrice, disPrice, finPrice
    char isMember;
    cout << "Please enter the purchase amount";
    cin >> purPrice;
    if(purPrice < 5000){
        disPrice = purPrice * (5/100);
    } else if ( purPrice >=-5000 && purPrice >15000){
        disPrice = purPrice * (10/100);
    } else{
        disPrice = purPrice * (15/100);
    }
    if (isMember==
    cout << "Enter Y for member";
    cin >> isMember;
    if(isMember == 'Y' || isMember == 'y'){
        disPrice = disPrice + purPrice * (5/100);
    }
    finPrice = purPrice - disPrice;
cout << "Original amount:" >> purPrice<<endl;
Cout << "Total discount:" >> disPrice<<endl;
cout << "Final amount:" >> finPrice<<endl;
if(finPrice>20000){
    cout << "You've earned a gold voucher";
}else {
    cout << "Thank you for shopping";
}

int arr[10];
```

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```
return 0;
}
```

5)
```
#include <iostream>
using namespace std;
int main(){
    int salary, creditScore, age;
    cout << "Enter your salary, credit score and age";
    cin >> salary;
    cin >> creditScore;
    cin >> age;
    cout << "Salary:" << salary << endl;
    cout << "Credit Score:" << creditScore << endl;
    cout << "Age:" << age << endl;
    if (salary >= 40000 && creditScore >= 700
        && age >= 21 && age <= 60){
        cout << "Loan approved";
    }
    else if (creditScore < 700 && salary > 60000
        && age >= 21 && age <= 60) {
        cout << "Conditional approval - Improve
            credit score";
        }
    else {
        cout << "Loan Rejected";
    }
return 0;
}
```